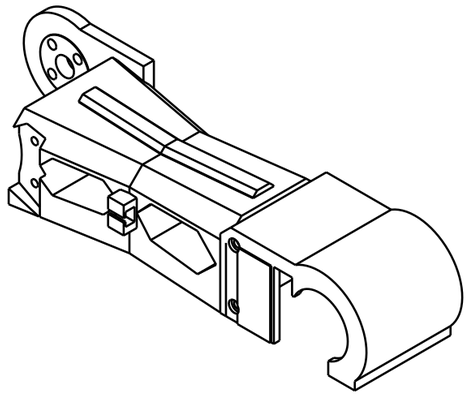
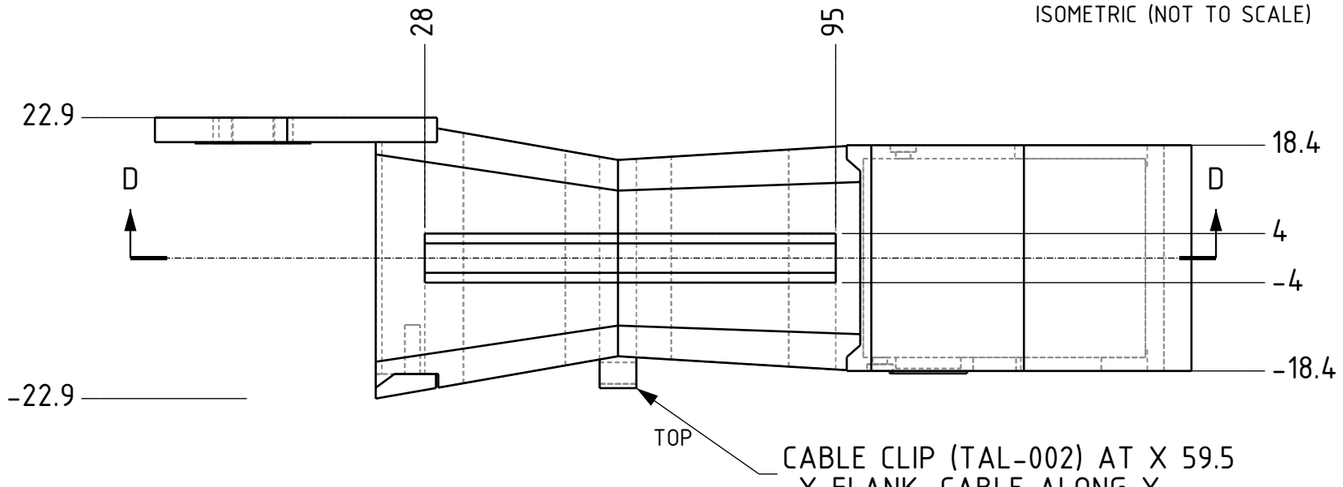
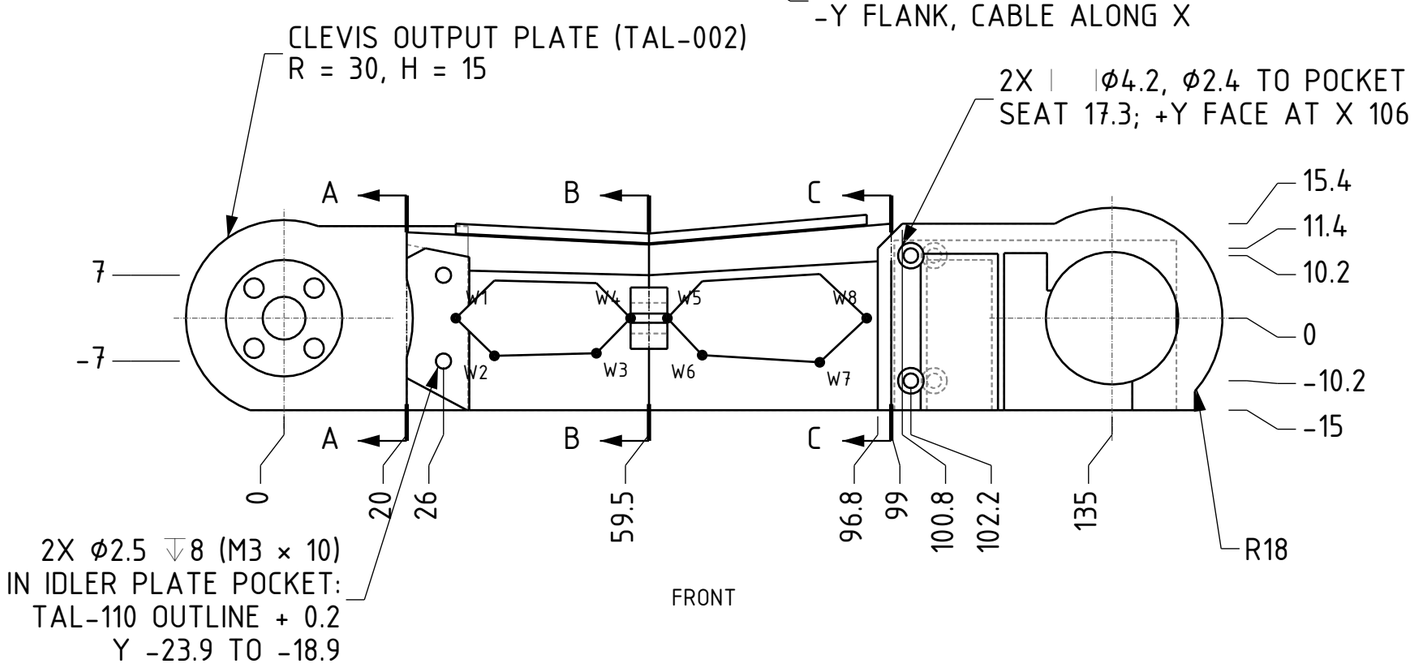


ISOMETRIC (NOT TO SCALE)




## WINDOW POINTS (X, Z), THROUGH Y

| PT | X | Z |
|---|---|---|
| W1 | 28 | 0 |
| W2 | 34.3 | ±6.14 |
| W3 | 50.92 | ±5.72 |
| W4 | 56.5 | 0 |
| W5 | 62.5 | 0 |
| W6 | 68.18 | ±6.03 |
| W7 | 87.34 | ±7.19 |
| W8 | 95 | 0 |

## FEATURES (in order)

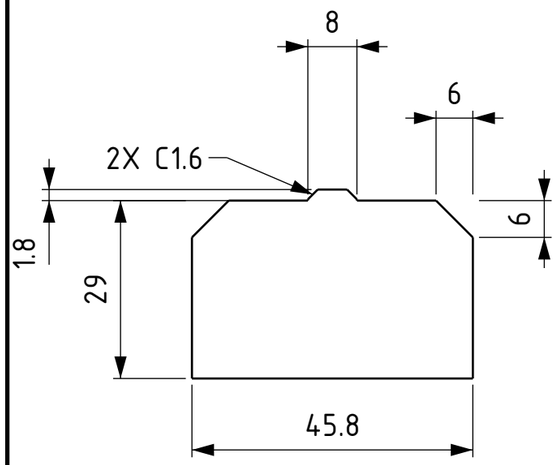
1. Clevis output plate only (TAL-002, +Y plate with its horn pad), R = 30, H = 15. The idler plate is the separate part TAL-110.
2. Beam: ruled loft of the profiles A, B, C (sheet 2) at x 20, 59.5, 99. Every profile sits on z -15, so the underside is flat; the waist dips on top only.
3. Servo shell: XZ polygon (96.8, -15) (135, -15) (135, 15.4) (100.8, 15.4) (96.8, 11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 135.
4. Flat underside: cut everything below z -15 (trims the clevis hub and the R18 cylinder).
5. Nose floor: box x 144 to 148.5, y ±18.4, z -15 to -11; ties the shell walls together on the bed (the servo entry cut later trims its back to x 145.5).
6. Doubler on the -Y wall: box x 103.8 to 116.4, y -18.8 to -18.4, z -15 to 10.5 (keeps the wall beside the back pad channel 0.8 thick; out of the next link's reach).
7. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 95.
8. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
9. Cable clip (TAL-002) on the -Y flank at x 59.5, cable along X.
10. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes; the output pad stays.
11. Pocket P (sheet TAL-001) with its servo frame origin P0 at (135, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along -Z.
12. Servo entry from below: cut box x 99.5 to 145.5, y ±16.2, z below 12.7 (opens the floor under the whole servo). In the -Y wall: back pad channel x 104.8 to 115.4, y -18 to -16.2, z below -9.5; notch through the wall x 117.4 to 138.3, z below 0 (joins the plug window and the idler hole).
13. Case screw holes (TAL-002, seat 17.3): +Y face at x 106, z ±10.2; -Y face at x 102.2, z ±10.2.
14. Idler plate pocket: the TAL-110 outline (circle R10.5 + neck N1 to N4) grown 0.2, cut from y -23.9 to -18.9.
15. Idler plate screw holes: 2X ⌀2.5 at x 26, z ±7, from y -18.9 8 deep (M3 × 10).

## NOTES

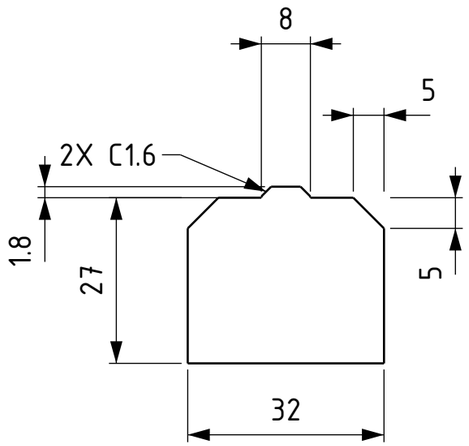
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the wrist flex servo from below, shaft to +Y, before its horns and plugs; then 4X M2 × 5 case screws.
- The wrist flex servo's bus plugs come out through the -Y shell wall.
- Print: flat underside down; no support (servo-opening roof and window tops bridge). The flat underside is z -15.

THIRD ANGLE PROJECTION

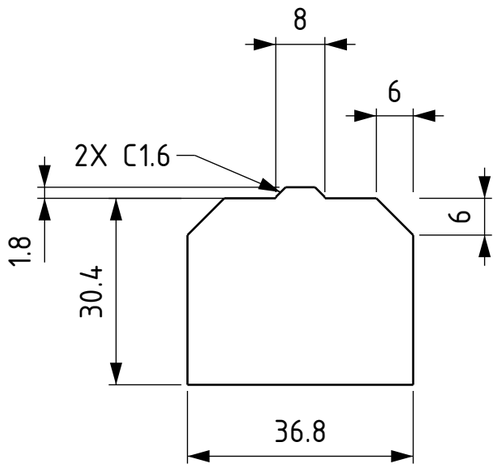
| WORK SET B: forearm, wrist, gripper, moving jaw, idler plates | |
|---|---|
| PROJECT<br>TALOS 5-axis arm | EST. MASS<br>54 g |
| PART<br>Forearm (part sheet 1 of 2) | SCALE<br>1:1 |
| PART NO.<br>TAL-104 | UNITS / REV<br>mm / A |
| QTY / ARM<br>1 | DATE<br>2026-11-07 |
| MATERIAL<br>PLA, 20 % infill, 3 perimeters | SHEET<br>12 of 21 |



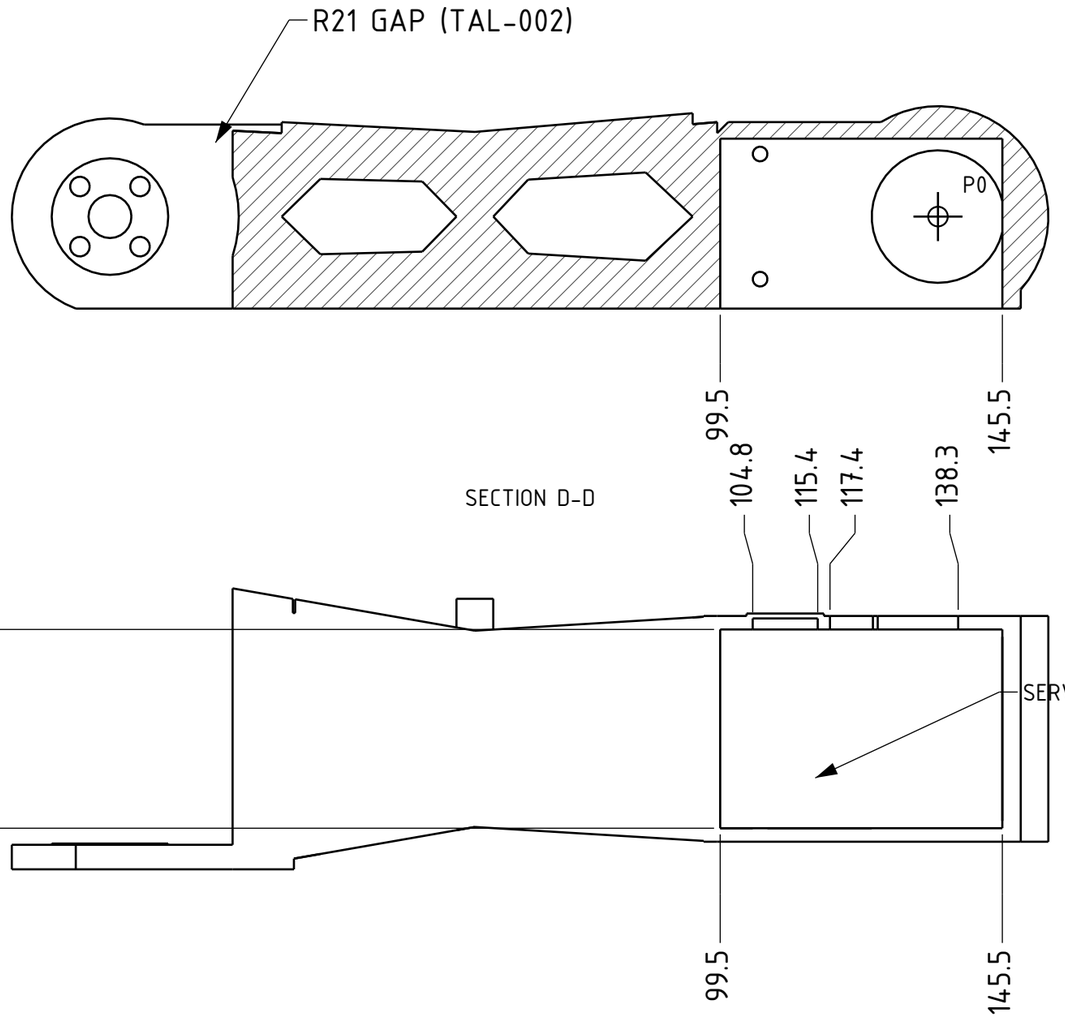

PROFILE A–A (X 20)




PROFILE B–B (X 59.5)




PROFILE C–C (X 99)




## PROFILES AND SECTION

Profiles A, B, C: beam section (top corners chamfered, underside flat and square-cornered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1. Each sits on z −15 (flat underside) with its top at the height given.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

Section D–D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

## NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

| WORK SET B: forearm, wrist, gripper, moving jaw, idler plates | |
|---|---|
| PROJECT<br>TALOS 5-axis arm | EST. MASS<br>– |
| PART<br>Forearm (part sheet 2 of 2) | SCALE<br>1:1 |
| PART NO.<br>TAL-104 | UNITS / REV<br>mm / A |
| QTY / ARM<br>1 | DATE<br>2026-11-07 |
| MATERIAL<br>PLA, 20 % infill, 3 perimeters | SHEET<br>13 of 21 |